SECTION 1

Time—30 minutes

45 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. ALLAY: (A) exacerbate (B) dismay (C) terrorize (D) invoke (E) blacken
2. ARTICULATE: (A) casual (B) obscure (C) incoherent (D) simpering (E) dull
3. OPAQUE: (A) prismatic (B) diaphanous (C) visible (D) refulgent (E) intangible
4. FETTER: (A) emancipate (B) rebel (C) escape (D) extend (E) revive
5. PROFICIENT: (A) careless (B) inexperienced (C) inappropriate (D) unskilled (E) hesitant
6. NONPARTISAN: (A) variegated (B) undivided (C) nominal (D) illicit (E) factional
7. ATROCIOUS: (A) fierce (B) admirable (C) normal (D) vivid (E) mottled
8. NULLIFY: (A) restate (B) notify (C) validate (D) promulgate (E) sign
9. PALATABLE: (A) culinary (B) garish (C) noxious (D) unflavored (E) bland
10. EMINENT: (A) unlikely (B) ill-favored (C) depressed (D) little-known (E) mythical
11. DESULTORY: (A) blank (B) sharp (C) pronounced (D) formal (E) determined
12. DUPLICITY: (A) simplicity (B) intelligence (C) incredulousness (D) unity (E) straightforwardness
13. BANAL: (A) brilliant (B) extraordinary (C) flagrant (D) notable (E) crucial
14. PRISTINE: (A) spoiled (B) vicious (C) faded (D) discolored (E) blended
15. CAPTIOUS: (A) yielding (B) indulgent (C) tactful (D) undesirable (E) indignant

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless..respectable (B) extensive..moderate (C) sophisticated..amateur (D) risqué..crude (E) perfect..spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

16. The songs of birds, like human languages, appear to develop ---- variations; for birds of the same species sing differently depending on their exact place of origin.

(A) familial (B) purposeful (C) genetic (D) numerous (E) geographic

17. Unlike the ---- pacifist, who regards nonviolence as a moral end in itself, the secular pacifist considers it simply one ---- of political struggle among many that may be adopted.

(A) sincere..method
(B) traditional..by-product
(C) conventional..goal
(D) religious..means
(E) spiritual..purpose

18. Marston's fine teaching is well known, but his ---- record of academic publication has led to ---- of his seriousness as a scholar.

(A) remarkable..consideration
(B) slender..notice
(C) dismal..praise
(D) scanty..disparagement
(E) extensive..questioning

19. The current oil glut should not obscure the fact that a worldwide depletion of oil reserves is ----; for world oil supplies are being used more ---- than they can possibly be replaced.

(A) likely..carelessly
(B) inevitable..quickly
(C) imminent..often
(D) improbable..rapidly
(E) unattainable..easily

GO ON TO THE NEXT PAGE

20. Neither anthropologist consciously distorted the facts in describing this remote tribe, but each ---- the available information in accordance with his particular ----.

(A) augmented..background
(B) revised..abilities
(C) interpreted..viewpoint
(D) twisted..bias
(E) altered..opinions

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

For at least 10,000 years—since long before the principles of classical genetics were scientifically established—human beings have brought about deliberate genetic changes in plants and animals through traditional reproductive methods. Many of the domestic animals, crops, and ornamental plants in existence today are human creations, achieved through selective breeding aimed at enhancing desired characteristics. In a broad sense, such genetic manipulation by breeding for a desired outcome might be considered "genetic engineering."

In addition to these intended changes, many alterations have occurred inadvertently through other practices, including the ordinary practice of medicine. Many people with genetic disorders who in the past would have died without any natural-born children now live into adulthood, passing on genes for their disorders. The use of insulin to treat diabetes and the prescription of eyeglasses for myopia are two examples of interventions that increase the prevalence in the population of certain genes that can have deleterious effects on individuals. Medical screening for genetic disorders, when leading to changes in reproductive behavior, also affects the occurrence of genes in the population. These changes have been by-products of medical and technological interventions aimed at individuals, not at the general population.

In 1965, the term *genetic engineering* was coined for what has come to be a wide range of techniques by which scientists can add genetically determined characteristics to cells that would not otherwise have possessed them. Compared with traditional means of altering the gene pool, the ability to alter genetic material directly offers specificity and, in the case of changes in germ cells, speed.

The rapidity with which this field has developed is startling. Scientists' understanding of the structure of deoxyribonucleic acid (DNA), which is common to almost all living cells, and their discovery of its remarkable capacity for encoding and passing on genetic characteristics are both relatively recent developments. In the early 1970s, scientists learned how to isolate specific DNA sequences from one species and attach this genetic material—*recombinant DNA*—to a different species. Rapid progress has also been made with cell fusion, another means of genetic engineering that permits the contents of two cells from different organisms to be merged in such a way that the hybrid cell continues to function and reproduce. These and other techniques hold the potential for producing dramatic changes in the genetic makeup of many species of plants and animals, including humans.

21. The passage is mainly concerned with

(A) clarifying the meaning of a scientific theory
(B) presenting evidence to support an unorthodox point of view
(C) summarizing the historical development of a scientific process
(D) refuting a widely held scientific misconception
(E) explaining the purposes of a technological process

22. According to the passage, the term *genetic engineering*, as coined in 1965, would include which of the following?

I. Medical screening for genetic disorders
II. Direct manipulation of cellular genetic materials
III. Genetic changes brought about through selective breeding

(A) I only
(B) II only
(C) I and II only
(D) I and III only
(E) I, II, and III

23. According to the passage, the earliest practitioners of selective breeding

(A) were able to produce only a highly limited number of genetic changes
(B) had no way of predicting the reproductive effects that their efforts would have
(C) were among the first to understand the structure of the genetic material in cells
(D) did not comprehend the genetic principles underlying their activities
(E) had no intention of bringing about the changes in species that they actually produced

24. The passage implies that the use of insulin to treat diabetes has resulted in

(A) an increase in the number of people genetically predisposed to diabetes
(B) a decrease in the survival rate of diabetic individuals
(C) the gene for diabetes becoming biologically useful rather than deleterious
(D) a decrease in the number of people considered diabetic
(E) an increase in the need for genetic screening of diabetic individuals

GO ON TO THE NEXT PAGE

25. It can be inferred from the passage that the development of recombinant DNA technology was most directly dependent on the

(A) rapid progress in cell fusion technology
(B) development of medical screening for genetic disorders
(C) coining of the term *genetic engineering*
(D) discovery of the structure and function of DNA
(E) improvement of traditional breeding techniques

The Iroquois developed the highest form of governmental organization of any Native American people north of Mexico. The Iroquois League was a loose confederation of member tribes with a council composed of fifty sachems. The sachems, whose titles belonged to various matrilineal families within the five tribes' several clans, usually held office for life.

Emphasizing both individual dignity and the power of the clan, the Iroquois League maintained local self-rule under a central government and developed a powerful defensive alliance. Though the league dealt with matters of war and foreign relations, it did not interfere in internal tribal affairs. In council, no action was taken without unanimous approval. It is believed that the writers of the U.S. Constitution were inspired by the league's highly developed sociopolitical organization and ideals.

Established around 1570, the Iroquois League immediately made its military power felt. It protected its members from attack by one another and gained a monopoly on the fur trade by overpowering such neighboring tribes as the Eries and Hurons. After acquiring firearms—first from the Dutch, then from the British—the league strengthened its power until its domain extended from Ottawa to the Tennessee River and as far west as Lake Michigan and the Illinois River. Despite losses in war, the league maintained its numerical strength by the Iroquois custom of adopting other tribes and remnants of tribes.

League warfare was intensified during the Colonial period. Iroquois leaders were astute diplomats, and armed Iroquois warriors became a force to be reckoned with in the struggle for the New World. Allied with the British during the French and Indian Wars, the league was decisive in breaking French power.

With the coming of the American Revolution, most Iroquois tribes continued their alliance with the British, partly due to the persuasion of Mohawk warrior Joseph Brant (Thayendanegea). However, the Oneidas and most of the Tuscaroras either fought on the American side or remained neutral. This split marked the end of the Iroquois League as a unified power.

26. The passage provides information to answer which of the following questions?

(A) How did the British win Iroquois support during the French and Indian Wars?
(B) Over what issue was the unity of the Iroquois League finally broken?
(C) What was the approximate population of the Iroquois League at the time of its greatest extent?
(D) How were the leaders of the individual Iroquois League tribes chosen?
(E) Why did the Oneidas side with the Americans during the Revolutionary War?

27. Which of the following political structures would probably most closely resemble the Iroquois League?

(A) An association of independent countries sharing some common administrative functions, such as postal service
(B) A federation of states under the control of a powerful centralized government
(C) A council of ministers from several countries charged with negotiating commercial agreements
(D) A unified nation subdivided for administrative purposes into many identically run districts
(E) A group of independent nations united by treaty for mutual self-defense

28. According to the passage, when a sachem died, his place on the council of the Iroquois League was taken by

(A) a successor elected by the remaining members of the council
(B) his eldest son
(C) a member of another tribe
(D) another member of his family
(E) the next oldest fellow tribesman

29. According to the passage, all of the following were characteristics of the Iroquois League EXCEPT

(A) a ban on internal warfare
(B) an emphasis on unity in decision making
(C) a system of balances among the various branches of government
(D) an aggressive foreign policy
(E) a representative form of self-government

30. The author's attitude toward the Iroquois League may best be described as one of

(A) romantic nostalgia
(B) approval and admiration
(C) profound skepticism
(D) disguised contempt
(E) balanced approval and disapproval

GO ON TO THE NEXT PAGE

Select the word or set of words that best completes each of the following sentences.

31. The city's aging and ---- water supply system has been increasingly subject to ---- in recent years.

(A) poorly maintained..breakdowns
(B) costly..improvements
(C) efficient..criticism
(D) extensive..enlargement
(E) highly regarded..accidents

32. Behavioral psychologists are reluctant to discuss any aspect of human activity that cannot be directly observed; as a result, they have little to say concerning individual ----.

(A) relationships (B) communication
(C) motivations (D) actions (E) behavior

33. While industrialization had blighted the neighboring towns and villages, East Henry had remained ----, a veritable rural ----.

(A) undeveloped..paradise
(B) peaceful..hamlet
(C) impoverished..backwater
(D) agricultural..shantytown
(E) uninhabited..desert

34. The classes are designed for a variety of participants ranging from the computer ---- to the sophisticated user in situations as ---- as elementary classrooms and national policymaking boards.

(A) illiterate..unusual
(B) enthusiast..various
(C) neophyte..peculiar
(D) amateur..numerous
(E) novice..diverse

35. Although the statement is ----, it is basically true that "Milton never repeated a form"; that is, he wrote only one significant work in each of the poetic ---- he attempted.

(A) false..modes
(B) oversimplified..genres
(C) surprising..creations
(D) obvious..styles
(E) exaggerated..eras

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN:BOREDOM:: (A) dream:sleep
(B) anger:madness (C) smile:amusement
(D) face:expression (E) impatience:rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

36. BASE:APEX:: (A) toe:heel
(B) foundation:structure (C) bottom:point
(D) foot:crown (E) abyss:slope

37. BARREN:FIELD:: (A) idle:labor
(B) colorless:landscape (C) dry:desert
(D) fruitless:orchard (E) sterile:offspring

38. GLANCE:EYE:: (A) sniff:nose
(B) swallow:throat (C) nibble:food
(D) comb:hair (E) eavesdrop:ear

39. GULLIBLE:FOOLISH::
(A) lazy:slow
(B) shrewd:unscrupulous
(C) innocent:cunning
(D) ingenuous:intelligent
(E) rational:sensible

40. REGIME:OVERTHROW::
(A) administration:oppose (B) royalty:behead
(C) military:suppress (D) president:elect
(E) incumbent:unseat

41. DELIBERATE:CHANCE:: (A) formless:chaos
(B) organized:disorder (C) random:sequence
(D) purposeful:reason
(E) meaningful:obscurity

42. HUNTER:POACH:: (A) hiker:trespass
(B) driver:crash (C) student:fail
(D) judge:condemn (E) archer:shoot

43. CONDOLENCE:GRIEF:: (A) emollient:burn
(B) sympathy:concern (C) relief:anxiety
(D) understanding:puzzlement
(E) joy:aggravation

44. TRACK:TIES:: (A) rail:spikes
(B) crossbar:uprights (C) dial:hands
(D) spoke:wheels (E) ladder:rungs

45. TAUNT:IRRITATION::
(A) denounce:disagreement
(B) praise:esteem
(C) degrade:humiliation
(D) tease:amusement
(E) embarrass:hatred

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 2

Time—30 minutes

25 QUESTIONS

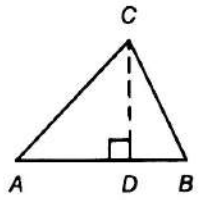
In this section solve each problem, using any available space on the page for scratchwork. Then indicate the one correct answer in the appropriate space on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definitions of symbols:

$=$	is equal to	$\leqq$	is less than or equal to
$\neq$	is unequal to	$\geqq$	is greater than or equal to
$<$	is less than	$\parallel$	is parallel to
$>$	is greater than	$\perp$	is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.
If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \dfrac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures which accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

---

1. If $\dfrac{7}{5} = \dfrac{4}{x}$, then $x =$

(A) $\dfrac{7}{20}$ (B) $\dfrac{4}{7}$ (C) $\dfrac{7}{5}$ (D) $\dfrac{20}{7}$ (E) $\dfrac{28}{5}$

2. If n is a positive integer, then which of the following must be divisible by 3?

(A) $9n - 6$ (B) $\dfrac{3n}{3}$ (C) $2n + 1$

(D) $3n - 1$ (E) n^3

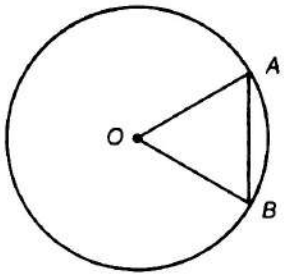
3. A store owner buys T-shirts for \$16.00 a dozen and sells them for \$3.50 each. If there are no other expenses, what will the total profit be on sales of 60 T-shirts?

(A) \$142.00 (B) \$130.00 (C) \$118.00

(D) \$106.00 (E) \$98.00

4. Four liters of ammonia are added to a tank of water, producing a cleaning solution that is 20% ammonia. How much more ammonia, in liters, must be added to increase the strength of the solution to 25%?

(A) 1 (B) $1\frac{1}{4}$ (C) $1\frac{1}{3}$ (D) $1\frac{1}{2}$ (E) 2



5. In circle O above, OAB is an equilateral triangle with side 9. What is the length of arc AB?

(A) 9π (B) 6π (C) 3π (D) $\dfrac{3\pi}{2}$ (E) π

---

0 1 0 0 1 1 0 0 0 1 0 0 0 0 1 1 0 0

6. The figure above represents a strip of paper on which 0s and 1s have been printed from left to right according to a certain pattern. If the pattern continues, the next three digits on the right will read

(A) 000 (B) 001 (C) 010

(D) 100 (E) 011

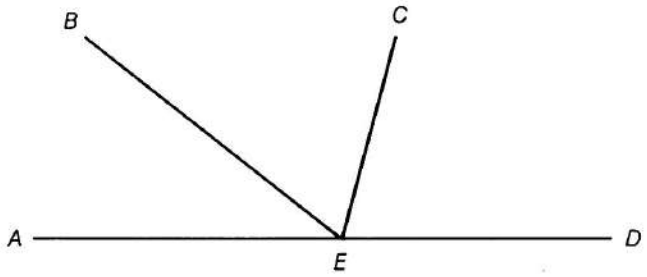
GO ON TO THE NEXT PAGE

7. If $2^6 - 2^x = 2^5$, then $x =$

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

8. What is the area of the largest possible circle that can be drawn inside a rectangle that has a length of 10 and a width of 6?

(A) 6π (B) 8π (C) 9π (D) 25π (E) 36π



9. In the figure above, AED is a line segment. If $\measuredangle AEC$ has a degree measure of 105 and $\measuredangle BED$ has a degree measure of 142, what is the degree measure of $\measuredangle BEC$?

(A) 58 (B) 60 (C) 63 (D) 67 (E) 71

Questions 10–11 refer to the following definition.

For any real number n, $\oplus n = \dfrac{n^2 + 1}{n + 1}$.

10. $\oplus 3 =$

(A) $\frac{5}{2}$ (B) 3 (C) $\frac{10}{3}$ (D) 5 (E) $\frac{4}{3}$

11. $\dfrac{\oplus 4}{\oplus 2} =$

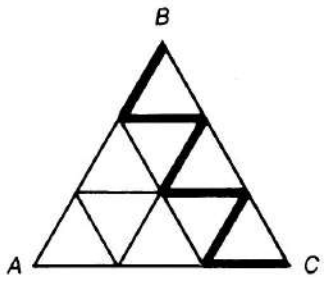
(A) $\frac{17}{5}$ (B) $\frac{51}{25}$ (C) $\frac{17}{25}$ (D) $\frac{32}{9}$ (E) $\frac{15}{17}$

12. If $x + 2 = 0$ and $y - 2 = 0$, then $\dfrac{x}{y} - \dfrac{y}{x} =$

(A) 2 (B) 0 (C) -1 (D) -2 (E) -4

13. Todd is twice as old as Alan. When Todd was 7 years older than Alan is now, he was three times as old as Alan. How old is Todd?

(A) 32 (B) 28 (C) 24 (D) 20 (E) 16



14. In the figure above, triangle ABC is an equilateral triangle with side 6 divided as shown into smaller equilateral triangles of equal size. What is the total length of the darkened path from vertex B to vertex C?

(A) 6 (B) 9 (C) 12 (D) 15 (E) 18

	Games Won	Games Lost
Team A	24	18
Team B	25	21

15. The chart above shows the records of two teams in a sports league. Team B will have won the same percentage of its games as Team A has won so far if Team B

(A) loses its next 2 games
(B) wins its next game
(C) wins 2 of its next 3 games
(D) loses its next game
(E) wins its next 3 games

16. If Bob works 8 hours at an hourly salary of d dollars and c cents, his total salary for the 8 hours will equal how many dollars?

(A) $\dfrac{8(d + c)}{100}$ (B) $8d + 800c$ (C) $8d + \dfrac{2c}{25}$

(D) $8d + \dfrac{c}{100}$ (E) $\dfrac{8d}{100} + 8c$

17. If $x = 4$ and $y = \dfrac{3x}{2}$, then which of the following equals 8?

(A) $2y - x$ (B) $\dfrac{2x}{2} + y$ (C) $\dfrac{x + y}{2}$

(D) $x + y$ (E) $\dfrac{y}{x} + 4$

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18. What is the volume in cubic inches of a rectangular solid with edges of $\frac{1}{2}$ foot, $\frac{1}{3}$ foot, and $\frac{1}{4}$ foot?

(A) $\frac{1}{2}$ (B) 6 (C) 24 (D) 36 (E) 72

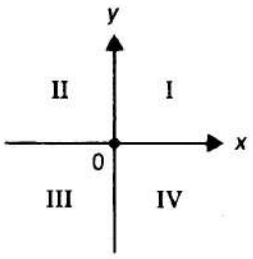
19. Two trains leave the same city and travel due east. Train A departs at 3 P.M. and travels at a rate of 50 miles per hour. Train B departs at 4:30 P.M. and overtakes Train A at exactly 7 P.M. the same day. What is the rate in miles per hour at which Train B is traveling?

(A) 65 (B) 70 (C) 80 (D) 86 (E) 90

20. If $x < y < 0$, then which of the following must have a value greater than 1?

(A) $x \times y$ (B) $x - y$ (C) $\frac{y}{x}$

(D) $y - x$ (E) $\frac{x}{y}$



21. In the figure above, which quadrants contain pairs (x, y) such that $x \times y$ is negative?

(A) II only (B) III only (C) I and III only
(D) II and IV only (E) I, II, III, and IV

22. Eight cards numbered 1 to 8 are in a pile. Two cards are drawn whose sum is 13. The remaining cards must include each of the following EXCEPT the card numbered

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

23. If $\frac{x}{2}$ is an odd integer, then which of the following must be an even integer?

(A) $\frac{x}{4}$ (B) $x + 1$ (C) $\left(\frac{x}{2}\right)^2$

(D) x (E) $x^2 - 1$

24. A school sells caps priced at \$3.50, \$5.00, and \$7.00. If it sells 54 caps priced at \$7.00 and collects a total of \$1,068.00 on sales of 210 caps, how many caps priced at \$5.00 does it sell?

(A) 60 (B) 78 (C) 90 (D) 96 (E) 104

25. If $4 - a < \frac{b+3}{2}$, then which of the following must be true?

(A) $a > \frac{5-b}{2}$

(B) $a < \frac{11-b}{2}$

(C) $a > 2b$

(D) $a > 4(b + 3)$

(E) $a < \frac{b-2}{2}$

S T O P

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SECTION 4

Time—30 minutes

40 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. HAPHAZARD: (A) continual (B) methodical (C) causal (D) repetitious (E) scientific

2. ANTAGONIZE: (A) sweeten (B) apologize (C) emulate (D) placate (E) hearken

3. STALWART: (A) apathetic (B) mild (C) feverish (D) craven (E) languid

4. RUE: (A) bless (B) recount (C) suggest (D) declare (E) please

5. MORIBUND: (A) bright (B) stable (C) vigorous (D) lighthearted (E) viable

6. EQUILIBRIUM: (A) strength (B) fissure (C) imbalance (D) movement (E) exuberance

7. LIMPID: (A) stagnant (B) swirling (C) erect (D) solid (E) turbid

8. COVET: (A) spurn (B) apply (C) cull (D) divest (E) abolish

9. IMPERVIOUS: (A) transparent (B) penetrable (C) indivisible (D) brittle (E) amenable

10. INDIGENT: (A) resident (B) fortunate (C) affluent (D) accepting (E) gracious

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless..respectable (B) extensive..moderate (C) sophisticated..amateur (D) risqué..crude (E) perfect..spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

11. Because these two primitive cultures are so widely separated, it seems unlikely that the similarities between them are due to ----.

(A) intercommunication (B) coincidence (C) error (D) proximity (E) chance

12. The human instinct for ---- cannot be regarded as an absolute, for many men and women have willingly sacrificed their lives for something they regarded as a greater good.

(A) survival (B) procreation (C) freedom (D) self-transcendence (E) happiness

13. Although baseball is often referred to as a "boys' game," the epithet is historically ----; the inventors and early players of the sport were all ----.

(A) accurate..amateurs
(B) sound..youthful
(C) dubious..professionals
(D) false..adults
(E) untenable..students

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14. Like Wallace and Darwin, who had developed the principles of evolution ----, Mondrian and Kandinsky reached the stage of pure abstraction in art by separate, unrelated paths.

(A) previously (B) accidentally
(C) together (D) analytically
(E) independently

15. The ---- comments of the professor eventually produced in Bramwell a fierce determination never to ---- in class unless it was absolutely unavoidable.

(A) helpful..remain
(B) vicious..listen
(C) measured..err
(D) jocular..appear
(E) carping..speak

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN:BOREDOM:: (A) dream:sleep
(B) anger:madness (C) smile:amusement
(D) face:expression (E) impatience:rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

16. MALIGN:LAUD:: (A) slander:libel
(B) condemn:praise (C) abhor:redeem
(D) extol:exalt (E) curse:defame

17. REDDEN:EMBARRASSMENT::
(A) blanch:horror
(B) blush:cynicism
(C) pale:illness
(D) shudder:anticipation
(E) marvel:awe

18. AIR:WING:: (A) motion:speed
(B) wind:propeller (C) ground:tail
(D) water:fin (E) jet:bird

19. CONSULT:DICTIONARY::
(A) examine:specimen
(B) measure:length
(C) gaze:stars
(D) write:paper
(E) peruse:scholar

20. LOCK:HAIR:: (A) weed:lawn (B) bone:ribs
(C) tuft:grass (D) leaf:tree (E) scale:fish

21. VIVID:HUE:: (A) icy:temperature
(B) shrill:speech (C) clarion:tone
(D) rounded:texture (E) blunt:point

22. IMMUNE:DISEASE:: (A) invisible:vision
(B) afflicted:suffering (C) curable:illness
(D) impregnable:attack
(E) vulnerable:defense

23. ANNULMENT:MARRIAGE::
(A) interruption:performance
(B) voiding:contract
(C) argument:conversation
(D) disagreement:friendship
(E) rejection:employment

24. FLUENT:GLIB:: (A) polished:slick
(B) plausible:deceitful (C) skillful:adroit
(D) accurate:precise (E) eloquent:loquacious

25. PEBBLE:GRAVEL:: (A) spoonful:sugar
(B) grain:sand (C) star:constellation
(D) captain:crew (E) stone:quarry

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

Heat is used to produce some of the gem tints rarely found in nature. Yellow topaz from Brazil, for example, can be heated carefully and cooled to produce a permanent pink, resembling the rare natural pink topaz. In a similar manner, a colored zircon may be heat-treated and thereby made colorless or changed to a different color.

Dyeing is another means of changing the natural color of a gemstone. This method is most often used on porous stones, such as the agate. Unattractive grays and pale shades can be changed chemically to bright red, green, and blue shades. The common dyeing procedure is to soak the agate in the desired chemical: nitric acid for red shades, nitrates and chromium salts for green shades, and certain cyanates and sulfates for blue shades. The Romans dyed the agate by soaking it in honey, removing it from the liquid, and heating it. This method produced black bands. Today the same effect is achieved by soaking the agate first in a sugar solution and then in sulfuric acid to char the sugar.

The ruby, spinel, emerald, sapphire, star sapphire, star ruby, rutile, and quartz are gemstones that have been successfully synthesized. A stone of simple chemical composition generally is more easily synthesized than one of complex composition. However, although the diamond has the simplest composition of any gem—it is pure crystalline carbon—synthetic gem diamonds have never been perfected.

GO ON TO THE NEXT PAGE

26. The author is primarily interested in describing

(A) the various colors in which gems occur in nature
(B) the effects of heat on different types of gem
(C) modern methods of improving the appearance of flawed gems
(D) artificial means of altering or imitating certain gems
(E) how chemical changes in gems may be produced

27. It can be inferred from the passage that gems are generally dyed

(A) for industrial purposes
(B) in chemical experimentation
(C) for use in optical procedures
(D) to reveal their internal structures
(E) for aesthetic reasons

28. The diamond is mentioned in the last sentence mainly as

(A) evidence for a theory
(B) an example of a common phenomenon
(C) an irrelevant digression
(D) an exception to a rule
(E) proof of a hypothesis

For nearly four centuries, western peoples accepted the division separating the "fine" arts of architecture, sculpture, and painting from the "lesser" arts. This boundary was fixed by the academies of art, and it was faithfully observed in education and in professional life, until its general abandonment about the time of the Second World War. It also marked off a class division between "upper-class taste" and "lack of taste" among the lower classes. The tacit abandonment of the rule of taste in this century thus corresponds to a wiping out of class distinctions, as well as of the separateness of the "fine arts" from "the others." But the study and the teaching of the history of art have not accepted this levelling of the subject. Historians of art stubbornly retain the criterion of "quality," and their attention is still drawn to "great" artists and to the "greatest" works in museums and collections. The objects of "low quality" made outside the rule of taste for proletarian consumers still are not accepted as proper subjects for searching art-historical study by university faculties and museum curators in either western or eastern Europe.

29. The author's attitude toward the distinction drawn between the fine arts and the lesser arts is one of

(A) tacit skepticism
(B) apologetic acceptance
(C) enthusiastic endorsement
(D) thoughtful resignation
(E) pompous disdain

30. The passage implies that art historians have devoted their attention mainly to works of art produced

(A) exclusively for display in museums and collections in western and eastern Europe
(B) by artists who are lacking in true aesthetic sensibility
(C) in art academies
(D) for an upper-class audience
(E) prior to the Second World War

31. With which of the following statements would the author be most likely to agree?

(A) Historians of art are to be commended for upholding standards of quality in an age of mediocrity.
(B) It is time for universities and museums to recognize the importance of the so-called "lesser" arts.
(C) The social distinction between the upper and lower classes is a result of the traditional distinction between the fine and lesser arts.
(D) The artistic perceptions of ordinary people generally lag behind those of scholars and educators in the world of art.
(E) The study of art would be improved if art historians learned to separate aesthetic judgments from commercial considerations.

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Ionizing radiation is applied in a number of useful ways in medicine. Low-energy X rays are used for diagnostic purposes. Generated by machine, these X rays are directed toward the part of a patient's body that a physician wishes to view. The X rays travel through the patient's body and transfer an image of the part exposed to a film behind the patient. By studying this film, specialists can detect a variety of abnormal conditions in the bone structure and the internal organs of a patient. Thus low-energy X rays aid physicians in diagnosing and prescribing suitable treatment for various internal disorders.

High-energy X rays, along with other forms of radiation, are used for therapeutic purposes, notably in the treatment of cancer. Although these X rays are harmful to all cells, they destroy cancer cells preferentially over normal cells. Normal cells appear to have a considerable capacity for repairing much, if not all, radiation-induced damage.

The effectiveness of radiation in destroying cancer cells—as well as the potential for harm to normal tissues from therapeutic radiation—depends on the type of radiation used and exactly how it is applied. Therapeutic X-radiation can be applied to part or all of the body. If it is applied to part of the body, a relatively strong dose can be administered. If it is applied to a larger area or to the whole body, the dose must be drastically reduced. Although scientists do not fully understand the relationship between volume of exposure and tolerance to radiation, they do know that whole-body tolerance is much lower than partial-body (or regional) tolerance.

Exposing a very large volume of the body to radiation is dangerous primarily because it may ultimately—years later—induce leukemia. Whole-body radiation has been known to cause this form of cancer, although the incidence of radiation-induced leukemia varies greatly with the level of exposure and the number of years of follow-up.

To avoid the possibility of inducing leukemia in patients, physicians usually treat cancer in a specific area of the body by subjecting the area to repeated applications of localized fields of high-level radiation for several days. The incidence of leukemia in persons receiving high-level partial-body X-radiation is no greater than it is in the population as a whole.

32. Which of the following titles best describes the content of the passage?

(A) The Dangers Posed by Radiation to the Human Body
(B) The Role of Radiation in the Diagnosis of Cancer
(C) The Medical Uses and Risks of Radiation
(D) Determining the Body's Tolerance for Radiation
(E) Radiation as a Major Cause of Cancer

33. According to the passage, radiation is medically useful because of its ability to do which of the following?

I. Travel through living cells
II. Create an image on film
III. Destroy living cells

(A) I only
(B) III only
(C) I and II only
(D) I and III only
(E) I, II, and III

34. According to the passage, high levels of radiation can be used with relative safety only

(A) in brief, widely spaced therapeutic doses
(B) when the radiation field is wide enough to include the entire body
(C) for a single intensive exposure
(D) when the cancer has spread over a large area of the body
(E) on strictly limited portions of the patient's body

35. The passage implies that one factor that makes the study of radiation-induced leukemia difficult is the

(A) uncertainty in diagnosing the disease
(B) lack of specialists trained in the treatment of the disease
(C) variation in the length of follow-up treatment provided
(D) ignorance of scientists about the levels of exposure likely to induce the disease
(E) length of time required for the disease to appear

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There are ultimately but three forces that control society—habit, coercive and violent force, and action directed by intelligence. In fairly normal
Line times, habit and custom are by far the strongest
(5) force. A social crisis means that this force has in large measure ceased to operate. The other forces, therefore, come more conspicuously into play. Reactionaries who strive to prevent any change of the old order are possessed of the power that
(10) enables them to use brute force in its less overt forms: by coercion, by intimidation, and by various forms of indirect pressure. From lack of understanding of social affairs, a lack of understanding owing to faulty education, as well as to deliberate
(15) refusal to learn, reactionaries unintelligently resist change. Those who have suffered from the old order then react by appeal to direct use of force as the only means at their command. Because of the intellectual suppressions experienced in the course of
(20) their own education they have little knowledge of means of effecting social changes by any method other than force.

In short, the social significance of academic freedom lies in the fact that without freedom of
(25) inquiry and freedom on the part of teachers and students to explore the forces at work in society and the means by which they may be directed, the habits of intelligent action that are necessary to the orderly development of society cannot be
(30) created. Training for good citizenship is one thing when conditions are simple and fairly stable. It is quite another thing when conditions are confused, complicated, and unsettled, when class divisions and struggles are imminent. Every force that
(35) operates to limit the freedom of education is a premium put upon ultimate recourse to violence to effect needed change. Every force that tends to liberate educational processes is a premium placed upon intelligent and orderly methods of
(40) directing to a more just, equitable, and humane end the social changes that are going on anyway.

36. The author defends academic freedom primarily on the basis of its

(A) benefits for scholarly activity
(B) protection by the Constitution
(C) importance for the advancement of science
(D) value to society as a whole
(E) moral and philosophical claims

37. It can be inferred that the author regards the major threat to academic freedom as deriving from

(A) reactionary forces seeking to suppress social change
(B) intellectuals wishing to inculcate their own political views
(C) revolutionaries attempting to indoctrinate students under the guise of education
(D) the atmosphere of chaos in education resulting from turmoil in the larger society
(E) economic forces that limit access to higher education

38. The author suggests that academic freedom is most necessary

(A) at times when most people are directed mainly by habit and custom
(B) during periods of social unrest
(C) when class distinctions have become solidified and rigid
(D) at times when society has attained a high degree of class stability
(E) when society is threatened by external enemies

39. Which of the following is an assumption made by the author?

(A) In today's world, a higher education is a prerequisite for effective citizenship.
(B) The use of force in directing social processes is never morally justified.
(C) It is impossible to arrest the process of social change.
(D) Those who oppose improvements in society often hide their beliefs behind the doctrine of academic freedom.
(E) The use of violence generally offers a strategic advantage to those wishing to suppress social justice.

40. This passage was most likely taken from

(A) a philosophical treatise on education
(B) an essay on the economics of social change
(C) a novel about academic freedom
(D) the editorial page of a reactionary student newspaper
(E) a speech advocating the direct use of force to effect needed changes in the public school system

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 5

Time—30 minutes

35 QUESTIONS

In this section solve each problem, using any available space on the page for scratchwork. Then indicate the one correct answer in the appropriate space on the answer sheet.

The following information is for your reference in solving some of the problems.

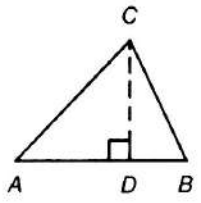
Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$

The number of degrees of arc in a circle is 360.

The measure in degrees of a straight angle is 180.

Definitions of symbols:

$=$	is equal to	$\leqq$	is less than or equal to
$\neq$	is unequal to	$\geqq$	is greater than or equal to
$<$	is less than	$\parallel$	is parallel to
$>$	is greater than	$\perp$	is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \dfrac{AB \times CD}{2}$

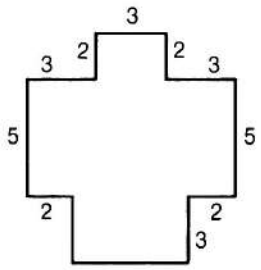
(2) $AC^2 = AD^2 + DC^2$

Note: Figures which accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

---

1. $\frac{1}{3}$ of $\frac{2}{5}$ of $\frac{3}{4}$ of $\frac{5}{6}$ of 36 equals

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6



2. In the figure above, all angles are right angles and the lengths are as shown. What is the total area of the figure?

(A) 64 (B) 66 (C) 70 (D) 75 (E) 82

---

3. If $x = \frac{y}{4}$ and $8x = 18$, then $y =$

(A) 9 (B) 8 (C) 7 (D) 5 (E) 4

4. If every even digit in the number 4321 is doubled, the value of the number will increase by

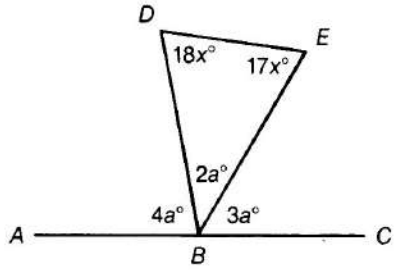
(A) 8642 (B) 8341 (C) 4321

(D) 4020 (E) 4000

5. Dates of the calendar are sometimes represented by digits. For example, March 10, 1985, is represented as 3-10-85. In this system, what month in 1986 includes the earliest date that is represented by five or six digits, three of which are the same?

(A) January (B) March (C) April

(D) June (E) August



6. In the figure above, ABC is a line segment. What is the degree measure of the largest angle of triangle BDE?

(A) 64 (B) 68 (C) 72 (D) 80 (E) 90

---

7. Both x and y are positive integers. If $(x + y)^2 = 64$ and $xy = 15$, then $x^2 + y^2 =$

(A) 24 (B) 28 (C) 32 (D) 34 (E) 44

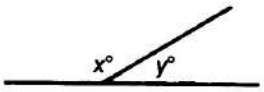
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Questions 8-27 each consist of two quantities, one in Column A and one in Column B. You are to compare the two quantities and on the answer sheet blacken space

A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

Notes: 1. In certain questions, information concerning one or both of the quantities to be compared is centered above the two columns.
2. In a given question, a symbol that appears in both columns represents the same thing in Column A as it does in Column B.
3. Letters such as x, n, and k stand for real numbers.

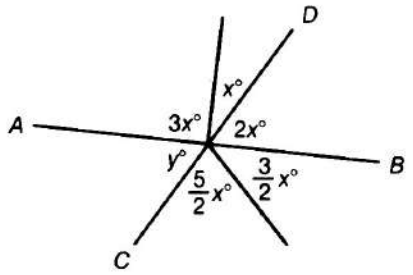
EXAMPLES

	Column A	Column B	Answers
E1.	2×6	$2 + 6$	● Ⓑ Ⓒ Ⓓ
E2.	 $180 - x$	y	Ⓐ Ⓑ ● Ⓓ
E3.	$p - q$	$q - p$	Ⓐ Ⓑ Ⓒ ●

	Column A	Column B
8.	$4 \times (3 + 3) \div 2$	$(4 \times 3) + (3 \div 2)$
9.	Number of centimeters in a kilometer	Number of milliliters in 100 liters
10.	$(2^3)^4$	$(2^6)^2$

$xyz = 18$
$x < 0$

11.	z	0

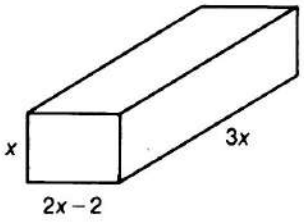


Line segment AB intersects line segment CD.

12.	y	60

$2x + 1 = 0$

13.	x^2	$\frac{1}{x^2}$

	Column A	Column B
		 $x = 3$
14.	Volume of the rectangular solid shown	100

Of 42 students in a class, $\frac{1}{3}$ have blue eyes and $\frac{1}{7}$ have green eyes. The rest have brown eyes.

15.	Number of students in the class with brown eyes	21

$a - b + c = 6$
$c - b - a = 4$

16.	a	2

$x \leqq y \leqq z$

17.	xy	yz

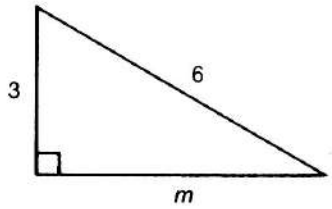
$\boxed{n}$ is defined as the smallest integer greater than or equal to n.

18.	$\boxed{3} + \boxed{-4.5}$	-1

GO ON TO THE NEXT PAGE

	Column A	Column B
	A box contains black marbles and white marbles. The number of black marbles is 3 more than the number of white marbles.	
19.	The chance of a blindfolded person choosing 2 black marbles consecutively	.50

	40% of $x = 18$	
20.	x	45

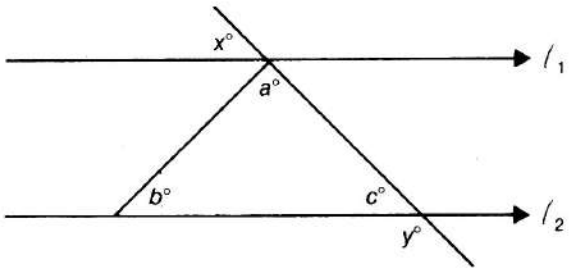


21.	m	5

Katia is n years old; Janet is 3 years older than Katia; Christopher is 4 years older than Janet.

22.	The sum of the ages of Katia, Janet, and Christopher	$3n + 10$

23.	Average (arithmetic mean) of the terms 4, 7, 6, 9, and 34	Average (arithmetic mean) of the terms 12, 12, 11, 13, and 11

	Column A	Column B
	 $\ell_1 \parallel \ell_2$	
24.	$a + b + c$	$x + y$

Triangle ABC is inscribed in circle O.

25.	Length of side AB	Diameter of circle O

$$\frac{x}{y} = 4$$

$$y = \frac{1}{4}$$

26.	x	1

$$x \neq 0$$

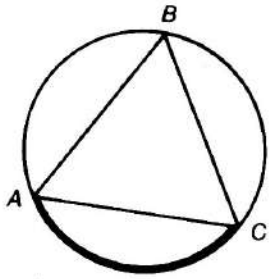
27.	$9x$	$\frac{3x}{2} \times 2x \div \frac{x}{3}$

GO ON TO THE NEXT PAGE

Solve each of the remaining problems in this section using any available space for scratchwork. Then indicate the one correct answer in the appropriate space on the answer sheet.

28. If $x = 3y + 5$ and $y + 3 = x - 10$, then $x + y =$

(A) 21 (B) 19 (C) 17 (D) 11 (E) 8



29. Equilateral triangle ABC is inscribed in the circle above. Point X (not shown) is located on the circle somewhere along darkened arc AC. If the circle has a radius of 15, what is the GREATEST possible length of arc BX?

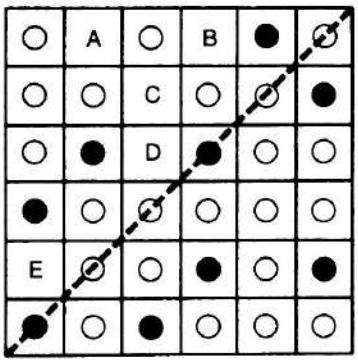
(A) 20π (B) 15π (C) 12π

(D) 10π (E) $7\frac{1}{2}\pi$

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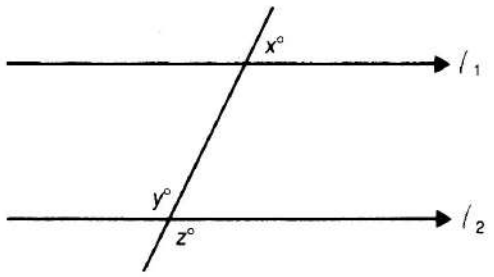
30. If $(x - 3) + (x - 2) + (x - 1) = 0$, then $3x =$

(A) 2 (B) 6 (C) $\frac{6}{4}$ (D) 9 (E) $\frac{9}{2}$



31. The grid above contains a pattern of black and white circles. If the pattern is symmetrical with respect to the dashed diagonal line, which of the lettered squares shown on the grid must contain a black circle?

(A) A (B) B (C) C (D) D (E) E



32. In the figure above, $\ell_1 \| \ell_2$. What is the value of $y + z$ in terms of x?

(A) $(90 - x) + (90 - x)$ (B) $180 - x$

(C) $90 + 2x$ (D) $180 + 2x$ (E) $360 - 2x$

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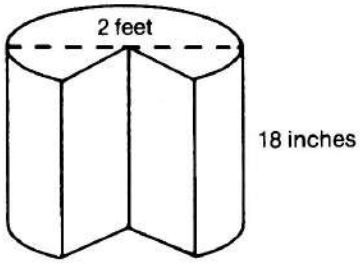
33. If $\frac{m}{n} = \frac{3}{4}$ and $3m = 18$, then $2n =$

(A) 8 (B) 10 (C) 12 (D) 14 (E) 16

34. In a coordinate graph system, point A is located at $(0, 2)$. If point B with coordinates (x, y) is equidistant from point A and the origin, then which of the following must be true?

(A) $x = 0$ (B) $y = 2$ (C) $y = 1$

(D) $x + y = 1$ (E) $y - x = 0$



35. The cylindrical wheel of cheese shown above has a diameter of 2 feet and a height of 18 inches. A wedge of cheese representing a quarter of the entire wheel is removed. What is the volume in cubic feet of the remaining piece of cheese?

(A) $\frac{3\pi}{2}$ (B) $\frac{18\pi}{7}$ (C) $\frac{9\pi}{8}$ (D) $\frac{2\pi}{3}$ (E) $\frac{4\pi}{5}$

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY. DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

Answer Key

SECTION 1

1. A	7. B	13. B	19. B	25. D	31. A	37. D	43. A
2. C	8. C	14. A	20. C	26. B	32. C	38. A	44. E
3. B	9. C	15. B	21. C	27. E	33. A	39. E	45. C
4. A	10. D	16. E	22. B	28. D	34. E	40. E	
5. D	11. E	17. D	23. D	29. C	35. B	41. B	
6. E	12. E	18. D	24. A	30. B	36. D	42. A	

SECTION 2

1. D	6. A	11. B	16. C	21. D
2. A	7. E	12. B	17. A	22. E
3. B	8. C	13. B	18. E	23. D
4. C	9. D	14. C	19. C	24. D
5. C	10. A	15. E	20. E	25. A

SECTION 4

1. B	6. C	11. A	16. B	21. C	26. D	31. B	36. D
2. D	7. E	12. A	17. A	22. D	27. E	32. C	37. A
3. D	8. A	13. D	18. D	23. B	28. D	33. E	38. B
4. A	9. B	14. E	19. A	24. A	29. A	34. E	39. C
5. C	10. C	15. E	20. C	25. B	30. D	35. E	40. A

SECTION 5

1. B	6. C	11. D	16. B	21. A	26. C	31. A
2. B	7. D	12. C	17. D	22. C	27. C	32. E
3. A	8. B	13. B	18. C	23. A	28. A	33. E
4. D	9. C	14. A	19. D	24. C	29. A	34. C
5. A	10. C	15. A	20. C	25. D	30. B	35. C

Score Conversion Table

Raw Score	SAT-Verbal	SAT-Mathematical
85	800	
84	780	
83	760	
82	750	
81	740	
80	730	
79	720	
78	710	
77	700	
76	690	
75	680	
74	670	
73	660	
72	650	
71	640	
70	630	
69	630	
68	620	
67	620	
66	610	
65	610	
64	600	
63	600	
62	590	
61	590	
60	580	800
59	580	780
58	570	770
57	570	760
56	560	750
55	550	740
54	550	730
53	540	720
52	540	710
51	530	700
50	520	690
49	520	680
48	510	670
47	500	660
46	500	650
45	490	640
44	490	630
43	480	620
42	470	610
41	470	600

Raw Score	SAT-Verbal	SAT-Mathematical
40	460	600
39	450	590
38	450	580
37	440	570
36	440	560
35	430	550
34	430	540
33	420	530
32	420	520
31	410	510
30	400	500
29	400	490
28	390	490
27	380	480
26	370	470
25	370	460
24	360	450
23	350	440
22	350	430
21	340	420
20	330	410
19	330	400
18	320	390
17	310	380
16	300	370
15	290	360
14	280	350
13	280	340
12	270	330
11	260	320
10	250	310
9	250	310
8	240	300
7	230	290
6	220	280
5	210	270
4	200	260
3	200	250
2	200	240
1	200	220
0 or less	200	200